

CSCE 581: Introduction to Trusted AI

Lectures 27-28: Class Project, Grad Paper Presentation

PROF. BIPLAV SRIVASTAVA, AI INSTITUTE

21AND 23 APRIL, 2026

Carolinian Creed: “I will practice personal and academic integrity.”

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Organization of Lectures 27, 28

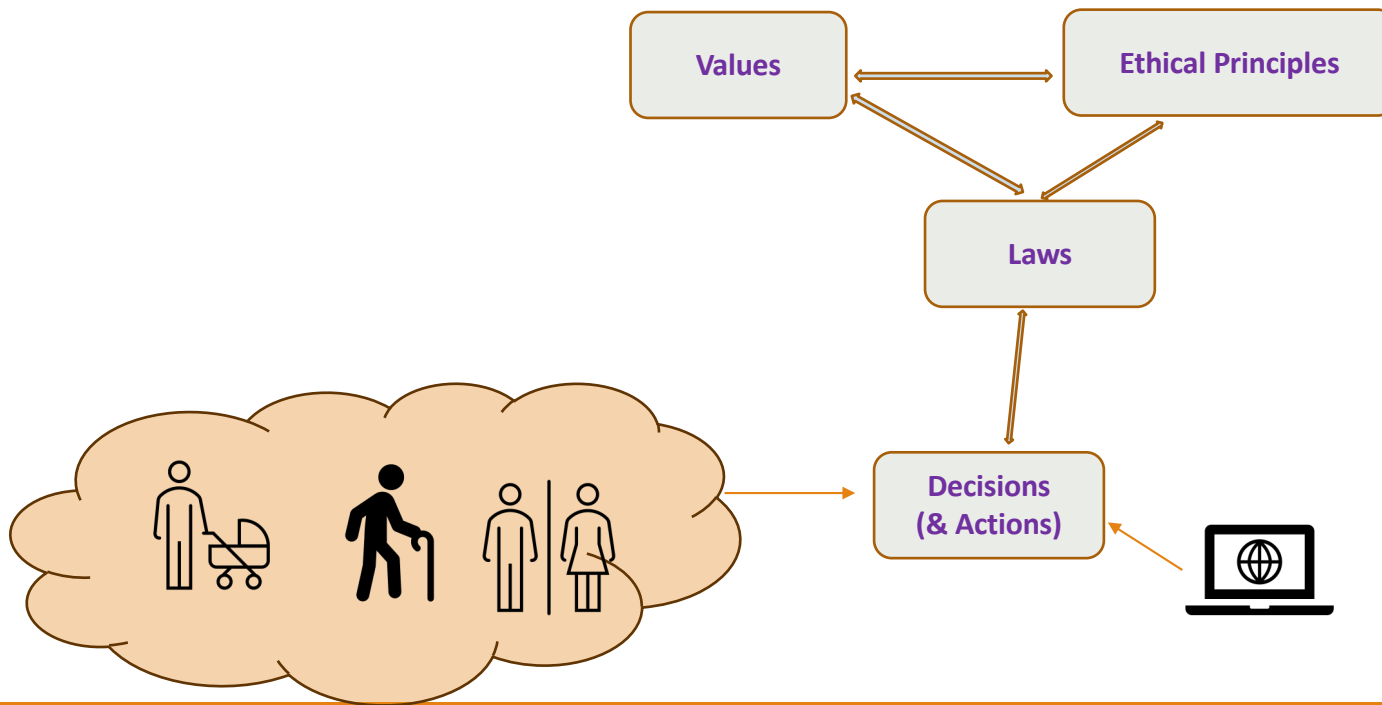
- Introduction Section
 - Recap from Week 13 (Lectures 25 and 26)
 - Announcements and News
- Main Section
 - L27: Project presentation
 - L28: Project presentation, (Graduate) paper presentation
- Concluding Section
 - About next week – Final/ May 5 - Lectures 30
 - Ask me anything

17	Mar 17 (Tu)	Invited Guest –	W10 Quiz 2 - start
18	Mar 19 (Th)	AI - Unstructured (Text): Processing and Representation	
19	Mar 24 (Tu)	AI - Unstructured (Text): Representation, Common NLP Tasks, Large Language Models (LLMs)	[Week 11] Quiz 2 - end
20	Mar 26 (Th)	Natural Languages/ Language Models and their Impact on AI	
21	Mar 31 (Tu)	AI - Unstructured (Text): Analysis – Supervised ML – Trust Issues	[Week 12] Project – Sprint 2 - end
22	Apr 2 (Th)	AI - Unstructured (Text): Analysis – Supervised ML – Mitigation Methods	Project – Sprint 3 - start
23	Apr 7 (Tu)	AI - Unstructured (Text): Analysis – Rating and Debiasing Methods	[Week 13] Quiz 3 - end
24	Apr 9 (Th)	Explanation Methods Trust: AI Testing	
25	Apr 14 (Tu)	Trust: Human-AI Collaboration	[Week 14] Quiz 3 - end
26	Apr 16 (Th)	Emerging Standards and Laws Trust: Data Privacy - Trusted AI for the Real World	
27	Apr 21 (Tu)	Project presentation	[Week 15] Project – Sprint 3 - end
28	Apr 23 (Th)	Project presentation	Last day of class
	Apr 28 (Tu)		Reading Day
29	April 30 (Th)	12:30pm – Final exam/ Overview	[Week 16] Optional, information shared

Recap from Week 13 (Lectures 25, 26)

- We looked at
 - L25: Human-AI Collaboration, Chatbots
 - L26: Agentic AI, Emerging AI Trust Landscape - Standards, Privacy

Values, Ethics, Law, Decisions (Actions) with Trust



Trust: *how can, humans or machines (AI) or in collaboration together, take decisions and actions following the same **values**, **ethical** principles and **laws** ?*

Assumption: *they should follow same values, principles and laws.*

Project Discussion

Project Milestones and Deliverables

- See course website for schedule and milestones:
<https://sites.google.com/site/biplavsrivastava/teaching/ai-csce-581-spring-2026-trusted-ai>
 - Sprints: 1 (Crawl) : Feb 5-24, 2(Walk) : Feb 24-Mar31 and 3 (Run): Apr 2-21
- Milestones – in-class presentations
 - Project discussion 1 – Feb 5
 - Project discussion 2 – March 5
 - **Project presentations 3 (final) – April 21 and 23**
 - **Project complete by April 30, 2026 in Github**
 - **Report, code, presentation (for completeness)**

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
- Timelines of reports [-10 points]
- Code missing [-10 points]
- Testcase missing [-10 points]

Rubric for Evaluation of Course Project

Project

- Project plan along framework introduced (7 points)
- Challenging nature of project
- Actual achievement
- Report
- Sharing of code

Presentation

- Motivation
- Coverage of related work
- Results and significance
- Handling of questions

Penalty

- Timelines of presentations [-10 points each miss; -30 total]
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Course Project

- **Framework**

1. (Problem) Think of a problem whose solution may benefit people (e.g., health, water, air, traffic, safety)
2. (User) Consider how the primary user (e.g., patient, traveler) may be solving the problem today
3. (AI Method) Think of what the solution will do to help the primary user
 1. Solution => ML task (e.g. classification), recommendation, text summarization, ...
 2. Use a foundation model (e.g., LLM-based) solution as the baseline
4. (Data) Explore the data for a solution to work
5. (Reliability: Testing) Think of the evaluation metric we should employ to establish that the solution will work? (e.g., 20% reduction in patient deaths)
6. (Holding Human Values) Discuss if there are fairness/bias, privacy issues?
7. (Human-AI) Finally, elaborate how you will explain the primary user that your solution is trustable to be used by them

Project Discussion: What to Focus on ?

- Problem: you should care about it
- Data: should be available
- Method: you need to be comfortable with it. Have at least two – one serves as baseline
- Trust issue
 - Due to Users
 - Diverse demographics
 - Diverse abilities
 - Multiple human languages
 - Or other impacts
- What one does to mitigate trust issue

Project Discussion

1. Create a private Github repository called “CSCE581-Spring2026-<studentname>-Repo”. Share with Instructor (biplav-s)
2. Create a folder called “Project”. Inside, create a text file called “ProjectPlan.md” (or “ProjectPlan.txt”) and have details by the next class (Jan 30, 2026)

1. Title:
2. Key idea: (2-3 lines)
3. Who will care when done:
4. Data need:
5. Methods:
6. Evaluation:
7. Users:
8. Trust issue:

<Project Title> - <Your Name>

Project Context

1. Title:
2. Key idea: (2-3 lines)
3. Who will care when done:
4. Data need:
5. Methods:
6. Evaluation:
7. Users:
8. Trust issue:

- Test Case – demonstrate working
 - E.g., <input, output, correct output – if different, trust observation>

1 min context, 2 min demo, 1 min expts, 1 min Q/A

<Project Title> - <Your Name>

Format for Final Presentation

Demonstrate effectiveness/ efficiency

- Metrics (F1, running time, ...)
- Empirical results
- Comparison with a LLM (why your method over a general alternative)

Conclusion

- Experience
- Q/A

1 min context, 2 min demo, 1 min expts, 1 min Q/A

Project Report

- Due by Tuesday, April 30, 2026
- Will contain:
 - Project context
 - Demonstration, including trust aspect. Potentially link to a video.
 - Experimental results: effectiveness, efficiency dimensions
 - **Related work (what most relevant prior work is out there)**
 - **Discussion: experience and how it may be extended**
 - Conclusion

Graduate Paper Presentations (23 Apr)

Presenters – Graduate Students

Audience - Undergraduates

- See paper presentation before class
- Hear all paper presentations
- Ask questions
 - How much you liked the presentation
 - What you liked about the paper
 - What you liked about the presentation

Concluding Section

Week 14 (L25 and 26): Concluding Comments

- We looked at project presentations
- Complete project report

About Next Week – Final Exam

Final Exam

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